# Multi-Aspect Sentiment Analysis System

## Automated Sentiment Analysis Report

## 1. Project Overview

This project analyzes Amazon product reviews using sentiment analysis, aspect extraction, emotion classification, trend analysis and customer segmentation. The system also provides business recommendations and negative sentiment alerts.

## 2. Dataset Summary

| Metric | Value |
|---|---|
| Aspect-level Records | 5717 |
| Customers | 3421 |
| Products in Current Dataset | 1 |
| Time Period | 2012 - 2014 |

## 3. Overall Sentiment Analysis

| Sentiment | Count |
|---|---|
| Positive | 4384 |
| Neutral | 1140 |
| Negative | 193 |

## 4. Aspect-wise Sentiment Analysis

| Aspect | Positive | Neutral | Negative |
|---|---|---|---|
| Battery | 40 | 13 | 3 |
| Camera | 947 | 313 | 55 |
| Customer_Service | 136 | 41 | 7 |
| Delivery | 220 | 43 | 9 |
| Design | 150 | 53 | 9 |
| Display | 37 | 15 | 2 |
| Performance | 1309 | 287 | 47 |
| Price | 965 | 175 | 26 |
| Quality | 217 | 59 | 11 |
| Size | 363 | 141 | 24 |

## 5. Eight-Emotion Analysis

| Emotion | Count |
|---|---|
| Joy | 2033 |
| No Emotion | 1659 |
| Fear | 403 |
| Trust | 390 |
| Anticipation | 246 |
| Sadness | 124 |
| Anger | 45 |
| Surprise | 14 |
| Disgust | 1 |

## 6. Customer Segmentation

| Customer Segment | Customers |
|---|---|
| Positive-Oriented | 2581 |
| Neutral-Oriented | 703 |
| Negative-Oriented | 137 |

## 7. Negative Sentiment Alerts

Total negative sentiment spike months detected: 6

| Month | Negative % | Alert |
|---|---|---|
| 2012-04 | 5.88% | Negative Spike |
| 2012-06 | 8.33% | Negative Spike |
| 2012-11 | 6.96% | Negative Spike |
| 2013-08 | 5.12% | Negative Spike |
| 2013-09 | 5.12% | Negative Spike |
| 2014-11 | 10.00% | Negative Spike |

## 8. Business Recommendations

**Battery**: Improve battery life and charging performance.

**Size**: Provide better size options based on customer needs.

**Design**: Improve product design and appearance.

**Camera**: Improve camera quality and image performance.

**Quality**: Improve product quality and durability.

**Customer_Service**: Improve customer support and response time.

**Display**: Improve display quality, brightness and resolution.

**Delivery**: Improve delivery speed, packaging and tracking.

**Performance**: Improve product speed and overall performance.

**Price**: Review pricing and provide better value for customers.

## 9. Project Limitations

The current review dataset contains reviews for only one product. Therefore, genuine competitor comparison cannot be performed without additional multi-product review data. Customer membership information is derived from review activity and should not be interpreted as actual Amazon membership data. The emotion classifier uses keyword-based lexical patterns.

## 10. Conclusion

The developed system provides multi-aspect sentiment analysis, sentiment trend monitoring, emotion classification, customer segmentation, business recommendations and negative sentiment alerts through an interactive dashboard. A Flask API also enables real-time review analysis.